

Acme Corp - Nexus Integration Hub

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Architecture Overview

System Overview

The Nexus Integration Hub is Acme Corp enterprise middleware connecting ERP (SAP), CRM (Salesforce), HRIS (Workday), and legacy mainframe systems. It handles event routing, schema transformation, and real-time sync orchestration across 40+ downstream systems.

Components

API Gateway (FastAPI) connects to Auth Service (OAuth 2.0 + SAML SSO) and Integration Engine (Celery workers). PostgreSQL stores metadata; Redis handles queues; Kafka handles event streaming.

API Gateway

- Framework: FastAPI 0.110+
- Authentication: OAuth 2.0 JWT bearer tokens (see authentication_guide.pdf)
- Rate limiting via Redis sliding window
- All endpoints return JSON; errors follow RFC 7807

Integration Engine

- Celery workers process sync jobs asynchronously
- Each job has states: pending, validating, running, completed, failed
- Rollback snapshots stored in S3 with 30-day retention
- Maximum concurrent jobs per tenant: 5

Data Stores

PostgreSQL 15.x — Job metadata, user sessions, audit log

Redis 7.x — Task queue, rate limit counters

Kafka 3.x — Event streaming between connectors

S3 — Sync artifacts, rollback snapshots

Deployment

Containerized via Docker; orchestrated on Kubernetes (Acme Corp EKS cluster).

Helm chart in deploy/helm/. CI/CD: GitHub Actions to staging to manual promote to production.

Health checks: /health (liveness), /ready (readiness).

Observability

Structured JSON logging, Prometheus /metrics, OpenTelemetry tracing (Jaeger backend).

Alerting: PagerDuty for P1, Slack for P2/P3.

API Versioning

Current version: v2. v1 deprecated as of 2025-01-15; sunset date 2025-07-01.